Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—**Artificial-intelligence (AI) data centers can provide demand response (DR), but a meter-only baseline cannot distinguish a realizable workload shift from strategic deferral, delayed rebound, or a transmission-feasible response. We present a security-aware workload-state verification and settlement framework that links the submitted job ledger, physical service state, and nodal network value. A sparse multi-period linear program enforces release, conservation, deadlines, capacity, migration, and cross-midnight completion. A validation-only convex selector and a two-sided pointwise credit band provide a test-independent false-credit and bounded-under-credit certificate. Signed dispatch-value differences, all-credible single-contingency (N–1) constraints, and exact coalition accounting then determine the value and its allocation. The study uses 5,188,507 BurstGPT requests, 68,664 tensor-window MIT Supercloud jobs and 71,128 full-horizon positive-energy jobs with measured graphics processing unit (GPU) telemetry, 54 temporally locked days, and public transmission systems. Against the matched feasible-quantile reference, unsupported credited response falls from 2.026218 to 1.910572 MWh/day (5.71%, paired three-day-block 95% interval [0.057, 0.534] MWh, Holm-adjusted $p = 0.0156$); normalized root-mean-square error (nRMSE) falls from 0.353737 to 0.348412 (1.51%, paired interval [0.0015, 0.0115], Holm-adjusted $p = 0.0105$). In the trace-anchored mechanism-isolation layer, complete N–1 exact settlement reduces error from \$192.69 to \$0.90/day across all 37 non-islanding IEEE Reliability Test System (RTS)-24 outages. This reduction is an evaluator-consistency diagnostic: when the workload verifier is scored end-to-end, the corresponding errors are \$207.67, \$155.25, and \$155.40/day for base-case exact, N–1 linear, and N–1 exact settlement, respectively. An independent real-execution replay covers every locked 15-minute slot, and a global job-level flow certificate exactly represents all 71,128 positive-energy joined jobs with maximum residual below 10^{-17} MWh. A continuous conversion-interval audit, nonlinear alternating-current (AC) checks across four public networks, a decision-time information-boundary audit, complete spatial permutations, and exact Shapley allocation provide separate evidence tiers for the claimed verification-to-payment chain. Because the public workload and transmission traces are not co-located with a utility event, the study is a trace-driven benchmark rather than a field intervention.**

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. Introduction

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value.

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice, despite known bias and manipulation incentives [1], [2]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [1]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral, dynamic capacity, and geographical load balancing can reduce energy cost and peak demand [3]. Geographical assignment provides the complementary spatial control mechanism [4]. Recent work assesses data-driven flexibility and aggregator coordination [5], [6], while multi-level dispatch refines delay-sensitive electrical response. Hierarchical optimal-consumption models further connect facility efficiency with response quality [7]. Virtual-link market clearing and space–time remuneration establish how geographical and temporal transfers can enter electricity markets [8], [9]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [10]. However, these methods optimize an operator-controlled schedule rather than verify an independently submitted DR counterfactual against an auditable workload ledger. This separation between computing feasibility and market verification motivates the auditable ledger used here.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [11], [12]; N–1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [13]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification rather than present them as one field intervention.

This paper makes three power-system contributions. 1) We formulate a provenance-bound workload-state demand model in which release, deadline, conservation, cross-midnight completion, capacity, and migration constraints are enforced simultaneously. The resulting job-level witness links each joined job to its immutable scheduler/DCGM record and preserves measured execution energy. 2) We construct a validation-only, continuous-arrival counterfactual verifier with a pointwise credit envelope. The output is an attainable spatial–temporal load trajectory rather than an unconstrained meter-only prediction, and its statistical performance is evaluated under an explicit false-credit/under-credit tradeoff. 3) We develop signed space–time nodal remuneration and a bilateral net-value contract under preventive N–1 dispatch, and extend the payment certificate from three finite telemetry quantiles to a held-out conversion interval by an endpoint theorem. The novelty is the coupling invariant that carries one committed workload state into physical verification and network valuation; the individual LP, dispatch, and allocation primitives are standard and are not claimed as new in isolation. The certificate is conditional on trusted scheduler and telemetry provenance: a cryptographic digest verifies record integrity after commitment but does not authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. System Model and Problem Formulation

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals. The measured ledger supplies arrival energy a_{skt} for source s, class k, and release interval t. The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d\in\mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline D_k, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$
$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end T^+ and final event-day release T^0, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0,T^+-D_k\}}, \tag{5}$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. This model follows workload deferral and geographical assignment foundations [3], [4]; its rolling implementation follows the state-retention principle in [10]. Waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, respectively. Because arrivals are fixed, replacing waiting duration by service time adds only a constant.

For provenance, each retained job is represented by the canonical tuple and its digest in

$$\begin{aligned}
\ell_j &= (\text{id}_j, t_j^{\text{submit}}, t_j^{\text{start}}, t_j^{\text{end}}, E_j, n_j^{\text{tel}}, q_j, \kappa_j, \sigma_j),\\
\mathcal{L} &= \text{sort}\{\ell_j : j \in \mathcal{J}\},\\
h_{\mathcal{L}} &= \text{SHA256}(\text{serialize}(\mathcal{L}))\,.
\end{aligned} \tag{6}$$

where E_j is the positive aggregated Data Center GPU Manager (DCGM) energy and n_j^{tel} is its raw telemetry-record count. The digest uses the Secure Hash Standard [14]; the canonical row order, one-to-one join, release–execution order, and raw-to-join energy equality are checked independently of the scheduling linear program (LP). Consequently, the flow witness certifies the exact committed ledger and measured profile, while the digest does not make an unverified claim about jobs that might have been submitted before the commitment. Experiment 16 reports this provenance certificate and reconciles the measured regional capacity envelope with the 118-MW nameplate that was committed before the validation/test split; the envelope is not used to select that nameplate.

The trust boundary is explicit. The verifier assumes that the scheduler and DCGM services are the authoritative sources for the fields used in ℓ_j, and that the digest is committed before the event decision. Under this assumption, a changed record is detectable and a joined record cannot be silently replaced after commitment. The digest does not, however, establish that a semantically false deadline, region label, service class, or energy value was truthful when it was signed. Authenticity of those fields requires an upstream scheduler attestation, meter signature, or trusted execution interface; it is outside the optimization certificate proved here.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p_{dt}^0 denotes the no-event counterfactual and p_{dt}^1 denotes the event trajectory generated by the declared workload optimization. The mechanism-isolation experiments use $p^{1,\mathrm{sim}}$ for this optimized response, whereas Experiment 13 uses $p^{1,\mathrm{meas}}$ for the independently measured MIT/DCGM execution trace. We reserve p^{obs} for measured execution used only as scoring truth; it is never supplied to an estimator or used to claim a causal utility event. For either response trace, $r_{dt} = p^0_{dt} - p^1_{dt}$ is signed: gross local response is $[r_{dt}]_+$, while net response retains negative rebound. For an estimate $\widehat{p}^0$, unsupported credited energy is

$$F = \Delta t \sum_{d,t\in\mathcal{E}} [\widehat{p}^0_{dt} - \max\{p^{\mathrm{obs}}_{dt}, p^1_{dt}\}]_+. \tag{7}$$

The audit computes (7) separately for every locked day so that a favorable aggregate error cannot hide unsupported positive credit. We also report normalized root-mean-square error (nRMSE), bias, precision, recall, F_1, under-credit, payment error, and overpayment.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q, and DR price π^{DR}. If reference day j adds event-window service δR_j, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\mathrm{DR}}}{H}\sum_{j=1}^{H}\delta R_j - \sum_{j=1}^{H}\Delta C_j. \tag{8}$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j. Linear-program sensitivity [15] gives the following exact local condition:

$$\frac{q\pi^{\mathrm{DR}}}{H} > \min_j c'_j \quad\Longleftrightarrow\quad \begin{array}{l}\exists \text{ a strictly profitable}\\ \text{local deviation.}\end{array} \tag{9}$$

The historical-baseline premise is documented in [1]; (8)–(9) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p^{\mathrm{native}}_t + Mp^{\mathrm{dc}}_t$ the nodal demand vector. For generator-to-bus map C_g, generator bounds, power transfer distribution factor (PTDF) H, and line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \ & \sum_i C_i(g_i) \\ \text{s.t. } & \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \le g \le \bar{g}, \\ & -\bar{f} \le H(C_g g - p) \le \bar{f}. \end{aligned} \tag{10}$$

The incidence mapping aggregates device demand at its connection bus, and (10) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{11}$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. Proposed Methodology

A. Framework and matched-information pre-estimation

Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting follow [16], [17]; extremely randomized trees follow [18]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days [19]. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same complete ledger is supplied to the post-event learner, the selected single projection, and the proposed convex verifier.

B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty ρ_ℓ, the first-stage program solves

$$\min_{x,y,z} \ \mathcal{C}_{\mathrm{w}}(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \ge \pm(p_{dt}(x) - \widetilde{p}_{dt}), \tag{12}$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [15]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here $\mathcal{C}_{\mathrm{w}}$ denotes workload operating cost; C_i denotes generator-segment cost in (10), and $V(p)$ denotes the network dispatch value. Thus (12) changes the objective, not the physical feasible set. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. Their validation-only convex combination $\bar{p} = \sum_\ell \alpha_\ell p^{(\ell)}$, $\alpha \ge 0$, $\mathbf{1}^\top\alpha = 1$, remains feasible because the common scheduling set is convex.

The coefficients minimize validation squared error subject to both total and daily-tail false-credit budgets, where conditional value at risk (CVaR) measures the latter. Let $\ell^\star$ be the single candidate with minimum worst contiguous-fold nRMSE and B^{ref}_j its validation false exposure on day j. For reserve $\beta \in (0, 1]$,

$$\sum_j F_j(\alpha) \le \beta \sum_j B^{\mathrm{ref}}_j, \tag{13}$$

$$\mathrm{CVaR}_{0.75}(F_j(\alpha)) \le \beta\mathrm{CVaR}_{0.75}(B^{\mathrm{ref}}_j). \tag{14}$$

The empirical conditional-value-at-risk epigraph follows [20]. Four contiguous folds select β; infeasible reserves cannot be selected. A second exact projection targets $\bar{p}$ while imposing, for every event sample,

$$p^{\mathrm{safe}}_{dt} \le p^{(\ell^\star)}_{dt}. \tag{15}$$

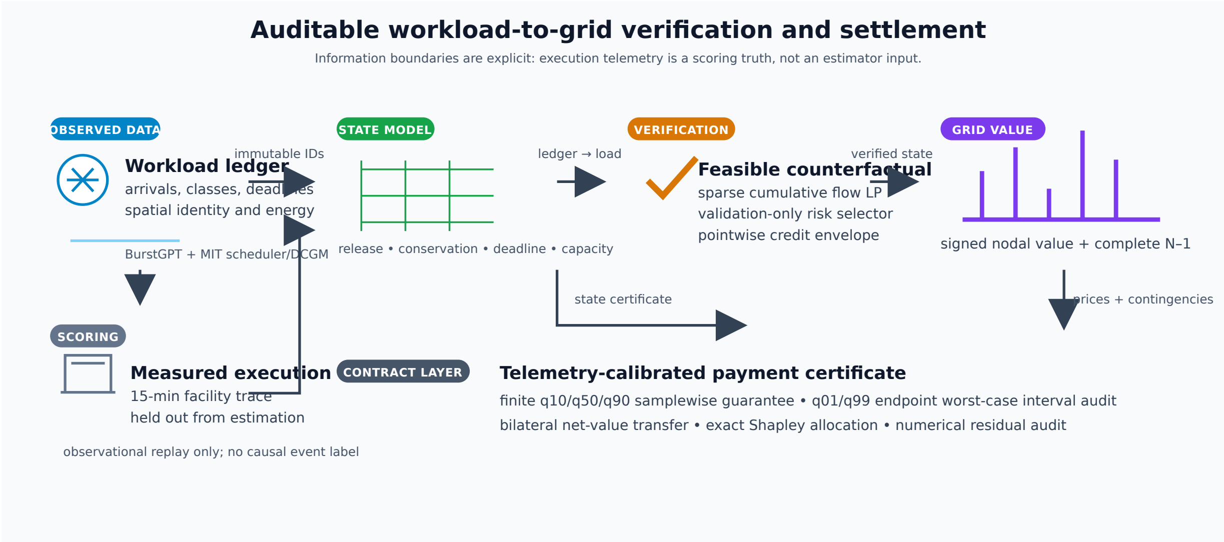

Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

a) Two-sided credit band.: The upper envelope alone would admit an arbitrarily conservative zero-credit profile. We therefore impose the matching lower band on the event slots,

$$p_{dt}^{(\ell^\star)} - \varepsilon \le p_{dt}^{\text{safe}} \le p_{dt}^{(\ell^\star)}, \qquad (d,t) \in \mathcal{D} \times \mathcal{E}, \tag{16}$$

where $\varepsilon = 1$ MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule while its credited event trajectory is within a declared band of the selected single projection. The total constraint (13) controls aggregate exposure and (14) controls the worst daily tail; the pair (15)–(16) supplies the samplewise credit certificate. Since $[z - c]_+$ is monotone, the upper inequality bounds false credit for any unseen actual trajectory, while the lower inequality bounds the additional under-credit by at most $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$ relative to the selected reference. The selected single feasible projection proves nonemptiness, and the locked results report both margins explicitly.

The cumulative state reduces the constraint-matrix nonzeros from $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$ for explicit prefixes to $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are solved to global optimality with HiGHS [21]. The resulting state-to-credit sequence is summarized in Fig. 2; the execution replay and interval audit are reported separately in Experiments 13–15.

C. Signed nodal and N–1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0 - p^1)$, while (11) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as an independent system operator (ISO) market payment. The convex subgradient inequality [15] yields the global certificate

$$0 \le (\lambda^0)^\top(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top(p^1 - p^0). \tag{17}$$

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space–time rule is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_{d} \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \tag{18}$$

This is the nodal virtual-link remuneration structure of [8], [9], applied here to a verified counterfactual. Summing site payments reproduces (18) exactly.

Space–time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^{\emptyset} - p_{dt}^1), \tag{19}$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \tag{20}$$

where $p^{\emptyset}$ is the minimum-cost no-event schedule under the identical continuous arrival window. Capacity credits and ex-post participation constraints follow established DR mechanism design [22], [23]; the signed Π^{ST} term separately debits spatial transfer and recovery. Equations (19)–(20) jointly define the signed service and operator value.

Let x^1 and $x^{\emptyset}$ denote the workload service schedules inducing p^1 and $p^{\emptyset}$, respectively. Define $C_{\text{opp}} = \mathcal{C}_{\text{w}}(x^1) - \mathcal{C}_{\text{w}}(x^{\emptyset})$ as the participant opportunity cost and $S = G - C_{\text{opp}}$ as the

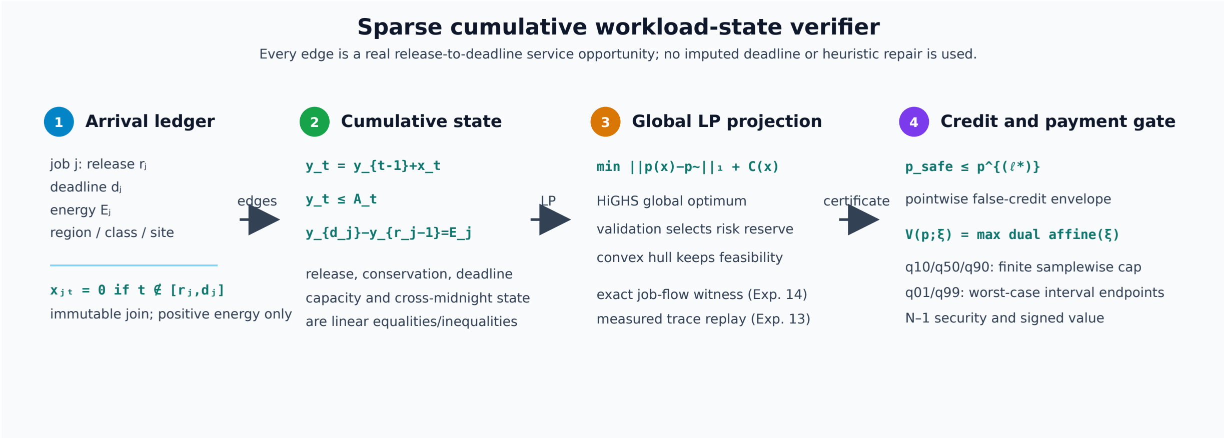

Fig. 2. Sparse cumulative workload-state verifier. Real release-to-deadline edges feed the global feasibility LP; the pointwise credit envelope and the finite/interval payment gates are applied only after the workload state is certified.

transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [24] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\mathrm{B}} = z\left(C_{\mathrm{opp}} + \frac{S}{2}\right), \quad u_{\mathrm{dc}} = u_{\mathrm{op}} = z\frac{S}{2}. \tag{21}$$

Thus voluntary nonactivation is the outside option rather than an omitted outcome. For every activated contract, both utilities are nonnegative and the single transfer P^{B} is exactly budget balanced. The bilateral transfer is reported separately from both ISO nodal remuneration and the exact dispatch value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t\in\mathcal{W}}(p^0_{dt} - p^1_{dt})\Delta t = 0$ up to solver tolerance.

Security-aware settlement augments (10) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \tag{22}$$

Equation (22) is the standard preventive N–1 LODF formulation [13]; the experiment enforces all credible rows simultaneously, rather than screening contingencies. Native public ratings are retained. Islanding outages are reported and excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the scope of DC security models and the N–1 planning criterion [25].

D. Scenario-robust N–1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \mathrm{conv}\{p^{(1)}, \ldots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\mathrm{ref}} \in \mathcal{P}$ be the validation-selected single feasible projection (the selected $\rho = 1$ member of the first-stage workload hull). The feasible-quantile projection is retained as an external matched transfer comparator in Section V. Let Ξ_3 contain the 10th, 50th, and 90th percentiles of held-out per-job measured-to-predicted GPU energy. These predeclared empirical scenarios define the finite power-conversion contract supported by independent telemetry [26]. For each settlement day, the payment-certified verifier defines $B^{\mathrm{ref}}_\xi = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p^{\mathrm{native}}_t + M\xi p^{\mathrm{ref}}_t)$ and solves

$$\begin{aligned}
\operatorname{lexmin}_{\alpha,e,\eta,\{g^\xi_t\}} \quad & \left(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\right) && (23)\\
\text{s.t.} \quad & p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)}, \\
& \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\
& e_{dt} \geq \pm(p^\alpha_{dt} - p^{\mathrm{safe}}_{dt}), \\
& g^\xi_t \text{ satisfies (10) and (22),} \\
& \quad \text{with } p_t = p^{\mathrm{native}}_t + M\xi p^\alpha_t, \\
& \qquad\qquad t \in \mathcal{E}, \quad \xi \in \Xi_3, \\
& \Delta t \sum_{t\in\mathcal{E}} \sum_i C_i(g^\xi_{it}) \\
& \quad \leq (1-\eta) B^{\mathrm{ref}}_\xi, \quad \xi \in \Xi_3.
\end{aligned}$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin η. Existence of the primal dispatches in (23) is equivalent to the corresponding upper bound on the optimal N–1 value; the unit weight on p^{ref} proves nonemptiness in every scenario. Consequently, for any $\xi \in \Xi_3$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\mathrm{native}}_t + M\xi p^\alpha_t) - V_{N-1}(p^{1,\xi}_t)] \\
&\quad \leq \sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\mathrm{native}}_t + M\xi p^{\mathrm{ref}}_t) - V_{N-1}(p^{1,\xi}_t)]. \qquad (24)
\end{aligned}$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-program epigraph foundations [12], [15]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (12) and the matched feasible-quantile projection. The risk-constrained verifier is an external target in the absolute-deviation objective and is not a selectable candidate; the quantile schedule is used only as an independent transfer comparator, not as the contractual payment cap.

E. Participant allocation

For participants N, define $v(S)$ as signed dispatch value when coalition S moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \tag{25}$$

This is the Shapley value [27]; the characteristic function based on signed grid value is the proposed specialization. Grouping (25) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\varnothing) = 0$. We enumerate all 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g (m_g + 1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

F. Analytical guarantees

Proposition 1 (manipulation boundary): Under a ten-day arithmetic baseline, differentiable one-sided optimal costs, and an interior feasible increase in at least one reference-day event service, a strictly profitable local manipulation exists if and only if (9) holds. To prove the statement, differentiate (8) along any feasible reference-day direction. The marginal payment is the same $q\pi^{\mathrm{DR}}/H$ for every day, whereas the least physical cost is $\min_j c'_j$. A strict excess selects a positive direction on an argmin day; if no strict excess exists, every nonnegative local direction has nonpositive first-order profit. Equality admits indifference but does not imply strict profit. Experiment 1 computes c'_j from the honest program's right-hand-side dual and solves the strategic program independently, preventing the behavioral result from being reused as its own certificate.

Proposition 2 (workload and two-sided credit feasibility): For fixed arrivals, every convex combination of solutions to (1)–(4) is feasible, and the envelope output satisfies the deterministic two-sided band (16) on every event cell. Consequently, for every possible unseen execution c_{dt}, its false-credit energy is no larger than that of $p^{(\ell^\star)}$, while its additional under-credit is bounded by $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$ relative to that reference. Affine equalities are preserved by the simplex coefficients; each linear inequality is preserved by convexity. The upper inequality in (16) and monotonicity of $[z-c]_+$ give

$$[p^{\mathrm{safe}}_{dt} - c_{dt}]_+ \le [p^{(\ell^\star)}_{dt} - c_{dt}]_+.$$

The lower inequality gives $[c_{dt} - p^{\mathrm{safe}}_{dt}]_+ \le [c_{dt} - p^{(\ell^\star)}_{dt}]_+ + \varepsilon$. Summation over sites and event intervals proves both bounds. The result is a feasibility-and-band certificate determined by the declared constraints and ε, not by the distribution of the locked test set.

Proposition 3 (security-aware value consistency): For the piecewise-linear base-case or N–1 dispatch, exact signed settlement equals the corresponding optimal-value difference and the nodal linear rule upper-bounds it by (17). The feasible-set right-hand side is affine in nodal demand, so the optimal value is convex and piecewise affine. Applying the subgradient inequality at p^0 gives (17); applying it again at p^1 yields

$$\begin{aligned} 0 \le \Delta V_{\mathrm{lin}} - \Delta V &\le (\lambda^1 - \lambda^0)^\top (p^1 - p^0) \\ &\le \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|. \end{aligned}$$

The same proof applies after stacking all rows in (22), because they remain linear. Gross positive-only settlement has no analogous identity: truncation changes the response vector before it enters the value function.

Proposition 4 (robust daily payment noninferiority): Every solution of (23) is workload feasible and its exact N–1 payment is no larger than the reference payment for every realized event trajectory and every declared $\xi \in \Xi_3$. Workload feasibility follows from Proposition 2. For each ξ, the embedded dispatch is a feasible witness whose cost satisfies the corresponding last constraint in (23); optimal N–1 dispatch can only have lower cost. Subtracting the same scenario-specific realized cost gives (24). Conditional on the telemetry-derived finite set Ξ_3, the proof is deterministic and contains no test-outcome assumption.

Proposition 5 (continuous worst-case interval certificate): For a fixed workload profile p, the optimal piecewise-linear N–1 dispatch value $V_{N-1}(p;\xi)$ is convex in a scalar conversion factor $\xi \in [\xi_{\mathrm{L}}, \xi_{\mathrm{U}}]$. Therefore

$$\max_{\xi\in[\xi_{\mathrm{L}},\xi_{\mathrm{U}}]} V_{N-1}(p;\xi) = \max\{V_{N-1}(p;\xi_{\mathrm{L}}), V_{N-1}(p;\xi_{\mathrm{U}})\}.$$

The value function is the supremum of affine dual objectives in ξ, hence convex; a convex function on a compact interval attains its maximum at an endpoint. If the two endpoint inequalities $V_{N-1}(p^\alpha;\xi_e) \le V_{N-1}(p^{\mathrm{ref}};\xi_e)$ hold for $e \in \{\mathrm{L},\mathrm{U}\}$, then the certified candidate also has no larger worst-case interval cost than the reference. This interval result is a worst-case bound; pointwise dominance at every interior ξ is asserted only for the finite set Ξ_3 in Proposition 4. Experiment 15 evaluates both endpoints with an independent 40-segment N–1 solver and reports this distinction explicitly.

Proposition 6 (provenance-bound workload witness): Suppose the canonical tuple in (6) is computed from the retained scheduler–DCGM join after the declared duplicate and positive-energy rules. If the certificate verifies one-to-one joining, release–execution ordering, raw-to-join energy conservation, and zero LP residuals, then the job-level flow witness is a feasible representation of that committed ledger. Under the collision-resistance assumption of SHA-256, any post-commit modification of a tuple or its canonical ordering changes the digest except with negligible probability. The statement binds the computational witness to the committed records; it does not certify that an uncommitted pre-event submission was truthful.

TABLE I
COMPLETE DATA AND EVALUATION PROTOCOL.

Item	Declared use and count
BurstGPT	5,188,507 requests; 2.176 billion tokens; inference arrivals and observed service
MIT scheduler	287,173 rows; submission time, duration, GPU count, and queue state
MIT GPU telemetry	96,893 rows; 68,664 tensor-window jobs and 71,128 full-horizon positive-energy jobs; measured execution energy
Time resolution	15 min; 121 calendar days; 121 valid complete days
Evaluation	16 validation days followed by 54 untouched test days
Networks	RTS-24, IEEE-39, Power Grid Library (PGLib)-118, and IEEE-300 with native ratings
Provenance	Immutable scheduler/DCGM join digest, raw-to-join energy conservation, and measured regional capacity envelope

Experiment 16 checks each premise independently and records the raw source hashes, so the provenance claim is separate from the scheduling solver's own feasibility status.

IV. Experimental Design

A. Data, preprocessing, and information boundary

Table I reports the complete data inventory. BurstGPT contributes 5,188,507 successful requests and observed token service [28]. MIT Supercloud contributes scheduler submissions and independent Data Center GPU Manager energy for 68,664 joined jobs [26]. Submission time creates queue arrivals; measured execution intervals create scoring truth. A held-out GPU-power regression uses 28,413 observations and obtains $R^2 = 0.899$, mean absolute error (MAE) 11.22 W, and root-mean-square error (RMSE) 15.82 W. Per-job measured-to-predicted energy ratios define Ξ_3 in (23); no synthetic row replaces an invalid record. The 68,664 count is the 121-day workload tensor used by Experiments 1–13. Experiment 14 does not truncate at that tensor boundary: it extends the deadline horizon to the last observed scheduler completion and retains all 71,128 positive-energy joined jobs for its independent job-level flow witness.

The public traces do not contain utility bus locations. Consequently, the four-region coupling is a trace-driven benchmark with fixed, predeclared bus placements and hyperscale conversion, not a utility field trial. Immutable row identifiers determine spatial partitions before any outcome is computed. This design permits exact repeated mechanism comparisons but does not identify a particular operator's geography. Public PGLib and MATPOWER/PYPOWER cases provide topology, capacities, costs, and ratings [12], [29].

B. Baselines, parameters, and inference

We compare High-5-of-10, ridge regression, gradient boosting, Extra Trees, online metadata gradient boosting, equally informed post-event metadata gradient boosting, complete-ledger quantile boosting, pre-event synthetic control, the exact feasible projection of the complete-ledger quantile, the tail-risk counterfactual, the independently selected single feasible projection, the risk-constrained verifier, and the payment-certified verifier. The first eight are meter/statistical comparators; the remaining methods receive the declared ledger as appropriate. An information-set audit verifies feature equality and confirms that no method observes execution truth.

Real-time inference has zero deferral, elastic inference four slots, and batch GPU work 512 slots. Per-site flexible capacity is 118 MW plus 6 MW fixed facility demand; the 118-MW nameplate is committed before the validation/test split and Experiment 16 only reconciles it with the complete observed MIT/DCGM batch envelope. Six projection penalties $\{0.03, 0.1, 0.3, 1, 3, 10\}$ and four risk reserves $\{0.60, 0.75, 0.90, 1.00\}$ are selected only on validation data. The settlement uses 10 generator-cost segments and is scored by an independently solved 40- or 80-segment evaluator. All 5,000 bootstrap replicates use moving three-day blocks [30]. Paired tests enumerate all 2^{18} signs of 18 nonoverlapping three-day blocks; Holm correction controls family-wise error [31].

The nonlinear model-out check uses the standard polar AC optimal power flow with active/reactive balance, voltage bounds, generator reactive bounds, and apparent-power ratings [12]. After solving the intact state, the preventive panel imposes, for every non-reference generator i and every credible outage k,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \qquad i \notin \mathcal{G}_{\text{ref}},\ k \in \mathcal{C}. \tag{26}$$

The reference generator balances only outage-dependent active losses; $Q_G^{(k)}$ and bus voltages are contingency recourse. Thus (26) tests one shared active plan rather than separately optimized post-contingency active schedules.

The evaluation follows this chain. Experiments 1–2 test the incentive model and workload-state verifier; Experiments 3–5 isolate settlement design and network transfer; Experiments 6–7 test active workload constraints and participant allocation. Experiment 8 evaluates all RTS-24 line outages, Experiment 9 tests the telemetry-robust payment certificate and freezes its candidate target and power scale on validation data, Experiment 10 transfers the counterfactuals to nonlinear AC models, and Experiment 11 exhausts all declared spatial assignments and power scales. Experiment 12 embeds each submitted event trajectory in a continuous real-arrival horizon and evaluates complete-cycle remuneration and bilateral participation. Experiment 13 replays the measured MIT/DCGM batch trace without intervening in the event, Experiment 14 solves one exact job-level flow LP for all joined jobs, and Experiment 15 independently audits the locked payment profile at the held-out q_{01} and q_{99} conversion endpoints. Experiment 16 audits immutable ledger provenance and reconciles the regional capacity with the same-source execution envelope. Experiment 17 fixes the information set at the event gate and compares a decision-time ledger with the complete-ledger audit. Experiment 18 evaluates AC post-contingency recourse across RTS-24, IEEE-30, IEEE-39, and IEEE-118 for every outage in a deterministic native-case admissibility set. Each confirmatory comparison uses the same 54 locked days unless a fixed network audit rule is stated; the case study is explanatory.

V. Results

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (9) agrees with the independently solved strategic program in all 56 cells. At $150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 4.09 MWh. Of 52.46 MWh credited, 18.05 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.344. The profitable region follows the derived marginal boundary rather than a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates all twelve methods on 54 later days. Table II separates the high-recall tail-risk counterfactual from the settlement-safe output. The tail-risk counterfactual improves mean credit F_1 from 0.485 to 0.513 over the selected single projection (Holm-adjusted $p = 0.0001$); its nRMSE is 0.353 versus 0.350, and the block test does not reject equality. The risk-constrained verifier combines seven independently feasible candidates without degenerating to one candidate. Relative to the equally informed feasible-quantile projection, unsupported credit falls from 2.026218 to 1.910572 MWh/day, a 5.71% reduction (paired three-day moving-block 95% interval [0.057200, 0.534363] MWh; Holm-adjusted $p = 0.015625$), while nRMSE falls from 0.353737 to 0.348412 (1.51%, interval [0.001547, 0.011528], Holm-adjusted $p = 0.010468$). The F_1 change is 0.485947 to 0.482752 (interval [-0.004470, 0.002890], Holm-adjusted $p = 1.000000$), so the result is a selective reduction in unsupported credit rather than a claim of uniformly better classification. The raw quantile learner exposes 87.223 MWh, showing that the principal gain is the auditable workload-feasible set and its deterministic credit envelope, not privileged execution truth. The same-ledger literature panel reports exact implementations of incentive-compatible spatial DR, price-responsive non-wire scheduling, proactive workload shifting, and aggregator coordination; its four 54-day means are retained in the baseline ledger rather than conflated with the proposed selector.

The two-sided certificate in (16) is checked independently for all 54 locked days and every event cell. With the predeclared 1-MW band, the mean false-credit exposure is 1.910572 MWh/day and the mean under-credit is 15.314000 MWh/day, compared with 15.244510 MWh/day for the selected single reference; every upper and lower pointwise margin is nonnegative. The RiskSafe therefore occupies a conservative point on an explicit false-credit/under-credit frontier: relative to feasible quantile it reduces false credit by 5.71% but increases under-credit by 0.089 MWh/day and does not improve F_1. The pointwise envelope prevents arbitrary downward collapse relative to the selected single profile, but it does not make the accuracy tradeoff disappear. We report this cost directly instead of treating the certificate as a uniform predictive improvement.

The closest domain baselines are implemented on the same locked ledger rather than represented by generic machine-learning surrogates. Their daily means are retained in the complete baseline panel: incentive-compatible spatial DR (nRMSE 2.108, false credit 89.846 MWh, $F_1 = 0.206$), price-responsive non-wire scheduling (0.361, 2.319 MWh, 0.444), proactive workload shifting (0.350, 1.957 MWh, 0.485), and aggregator coordination (0.552, 0 MWh, 0). The incentive-compatible and flexibility formulations follow [5], [23]; the receding-horizon and aggregator implementations follow [6], [10]. The complete day-level rows are retained with the released experiment outputs.

TABLE II
Locked 54-day counterfactual results (daily means).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	2.653	118.820	7.496	0.230
Ridge	1.845	90.582	4.820	0.321
Gradient boosting	2.192	103.378	7.434	0.256
Extra Trees	2.173	100.920	7.247	0.263
Online metadata	2.123	98.681	6.795	0.275
Post-event metadata	1.968	91.107	6.610	0.299
Post-event quantile	1.891	87.223	6.579	0.324
Synthetic control	1.893	88.185	9.163	0.270
Feasible quantile	0.354	2.026	15.225	0.486
Single feasible	**0.350**	1.957	15.245	0.485
Tail-risk feasible	0.353	4.180	13.153	**0.513**
Risk-constrained safe	**0.348**	**1.911**	15.314	0.483

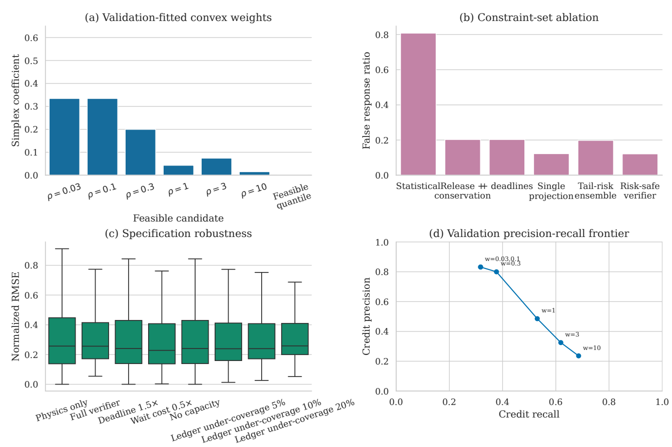

Fig. 3. Validation-fitted projection weights, constraint-set ablation, specification robustness, and the precision–recall frontier. Every point is produced by a globally solved workload program.

Fig. 3 reports the complete day-level distributions, constraint ablation, validation frontier, and independent intervention robustness. The 5-by-3 capacity/deadline factorial contains 810 optimal schedules; maximum capacity- and deadline-binding shares are 0.310 and 0.948. The measured nonzero effects establish that the associated constraints are not decorative. Matrix nonzeros scale with log–log slope 1.003 over 96–608 intervals, matching the linear-horizon proposition.

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table III first uses the trace-anchored reference to isolate the mechanism. Exact signed nodal value has mean absolute error $2.34, versus $741.70 for nodal gross response. Under the workload verifier, the errors are $523.25 and $632.11, respectively; the remaining error is dominated by counterfactual uncertainty rather than value-function linearization. The paired one-factor decomposition

TABLE III
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Uniform gross	741.72	632.58	–	–
Base case	Nodal gross	741.70	632.11	–	–
Base case	Uniform signed	6.01	524.05	–	–
Base case	Nodal signed linear	6.00	524.23	–	–
Base case	Nodal exact net value	**2.34**	**523.25**	–	–
N–1	Base exact	192.69	207.67	98.50/185.21	126.36/156.89
N–1	N–1 linear	2.30	**155.25**	0.61/2.08	111.83/37.53
N–1	N–1 exact	**0.90**	155.40	0.82/0.01	112.62/36.97

prevents attributing this bundled improvement to location alone. With the trace reference, signed netting lowers error by \$735.69/day (Holm-adjusted exact block-sign $p = 3.1 \times 10^{-5}$), whereas changing uniform signed prices to nodal prices changes it by only \$0.0062/day ($p = 0.0874$), and replacing nodal linear value by exact value lowers it by \$3.66/day ($p = 0.0874$). Under the workload verifier, the corresponding effects are \$109.34, -\$0.182, and \$0.988/day, none significant after Holm correction ($p = 0.706$). Thus signed rebound accounting identifies the dominant base-case gain; locational and security effects require the congested and N–1 panels below. All 5,616 interval/counterfactual instances satisfy (17); the largest numerical violation is 2.03×10^{-11} dollars.

Experiment 4 applies a predeclared rule—the largest gross-minus-net spatial offset—and selects day 69. The source curtailment, remote rebound, nodal prices, and line utilization are retained in the experiment output. The case explains why positive-only local accounting fails: migration creates a simultaneous negative component that gross payment discards.

C. Cross-network and N–1 security evidence

Experiment 5 contains 6,480 outcomes: four public networks, three native-load multipliers, two counterfactual qualities, five mechanisms, and all 54 days. No branch is derated and all data-center buses are fixed before outcomes are observed. Congestion arises endogenously in three of 12 network-load cells. In the trace-anchored mechanism-isolation layer, exact value reduces absolute error relative to signed-linear settlement in all 12 cells, with a mean improvement of \$1.92/day; ten of the twelve Holm-adjusted temporal-block tests are below 0.05. In the end-to-end workload-verifier layer, exact value improves five of the twelve cells, while the seven remaining differences have the opposite sign; the across-cell mean improvement is only \$0.016/day. This separation retains the global value-consistency claim of Proposition 3 without treating a trace-anchored mechanism effect as end-to-end baseline dominance or attributing counterfactual estimation error to the settlement rule.

The near-affine RTS-24, 0.98-load boundary cell is additionally recomputed on all 54 days at settlement/evaluation segment pairs 10/40, 20/80, 40/160, and 80/320. Its mean signed-linear-minus-exact error differences are -0.01695, -0.02730, -0.01938, and -0.01740 USD/day, respectively; the four differences remain below 0.03 USD/day in magnitude. This convergence identifies a locally affine dispatch region rather than a favorable discretization. The end-to-end cell effects, mechanism-isolation errors, and resolution convergence are retained in the experiment output.

Experiment 6 tests whether the workload constraints remain consequential when physical slack is changed. The complete factorial crosses capacity multipliers $0.55, 0.70, 0.85, 1.00, 1.20$ with deadline multipliers $0.25, 0.50, 1.00$ over all 54 locked days, giving 810 globally solved schedules. Capacity-binding share reaches 0.0411 and deadline-binding share 0.9118; the corresponding mean false-credit exposures are 2.641, 2.225, 1.971, 1.832, and 1.941 MWh across the capacity levels. Every retained schedule satisfies the release, deadline, conservation, and capacity constraints, so the stress panel measures sensitivity of the verifier rather than feasibility repair.

Experiment 7 evaluates exact budget-balanced allocation. The four-participant panel enumerates all 16 coalitions for every locked day and compares exact Shapley net value with leave-one-out, standalone avoided-cost, and signed-linear allocations. For the trace reference, exact Shapley has zero mean participant error and a maximum budget residual below 2.3×10^{-13} dollars; the workload-verifier error is 107.28 while the exact budget residual remains 1.6×10^{-13} dollars. The eight-participant panel enumerates all 256 coalitions and reports the signed site-level values, and the symmetric scaling panel reaches 20 participants with 1,296 evaluated states and maximum budget residual 2.98×10^{-13} dollars. The complete constraint-stress and allocation panels are retained in the experiment output.

Experiment 8 closes the security gap by enforcing (22) for every finite non-islanding outage in RTS-24 [32]. The public case has 38 lines: all 37 non-islanding single-line outages are included and the one islanding outage is explicitly reported. The full panel contains 2,592 interval-mechanism outcomes and 324 daily outcomes over the same 54 locked days. Native ratings, 0.90 native load, 6% peak data-center penetration, and four fixed sites are held constant. A 10-segment security-aware settlement is scored by a separate 40-segment N–1 dispatch. Fig. **??** compares base-case exact, N–1 linear, and N–1 exact value. With the trace-anchored counterfactual, their mean daily absolute errors are \$192.69, \$2.30, and \$0.90, respectively. The N–1 exact rule improves on base-case exact by \$191.79/day (54-day three-day-block exact sign test, Holm-adjusted $p = 3.1 \times 10^{-5}$); its additional \$1.40/day reduction relative to N–1 linear is not significant after Holm correction ($p = 0.122$). Under the workload verifier, the corresponding errors are \$207.67, \$155.25, and \$155.40; the linear-to-exact difference is \$0.14/day and nonsignificant ($p = 0.690$), showing that counterfactual error, not value linearization, is then dominant. The independent evaluator finds maximum post-contingency utilization 1.000 p.u. in every retained outcome. Table III retains both counterfactual qualities rather than reporting only the mechanism-isolation reference. The overpayment column is signed: positive values indicate payment above independently evaluated security value.

Experiment 9 tests the finite-scenario payment claim in (24). All 54 daily lexicographic program pairs attain global optimality. The candidate hull contains six first-stage projections and the matched feasible-quantile comparator, while the contractual payment cap is anchored to the validation- selected single feasible projection; the risk-constrained verifier remains an external target. The held-out conversion factors are 0.734, 0.996, and 1.278, which form Ξ_3. Across all $54 \times 3 = 162$ daily scenario certificates, maximum positive cap violation is zero and mean cap margins are \$0.596, \$0.809, and \$1.038/day. The payment-target candidate is selected as $\rho = 1$ by an independent validation-only N–1 payment-MAE panel containing 384 cells; neither this selection nor the power scale uses locked-test payment or peak outcomes. Independent 40-segment mean absolute errors for the payment-certified verifier are reported against both the selected single projection and the external feasible-quantile transfer comparator. Thus the independent finer-cost evaluator need not inherit the within-model error ordering, while (24) preserves the minimum trajectory deviation and certifies the declared-market samplewise cap relative to the selected single projection; averaged over the held-out q_{10}, q_{50}, and q_{90} scenarios, the corresponding errors are \$156.370/day for the payment-certified verifier, \$155.605/day for the selected single projection, and \$155.703/day for the feasible-quantile comparator (the risk-constrained verifier is \$155.562/day). These values are independent scoring results, not replacements for the deterministic cap. The maximum post-contingency loading remains 1.000 p.u. Experiment 15 separately audits the closed held-out conversion interval and is interpreted through Proposition 5. The cap is therefore a contractual upper bound under the declared market model; it is not evidence that the payment-certified profile has lower independent prediction error than every comparator.

Experiment 10 replaces the linear network with nonlinear AC optimal power flow [12]. Its 864 outcomes cross RTS-24, IEEE-30, IEEE-39, and IEEE-118 over 54 days and four workload-feasible counterfactuals. Native active/reactive loads, 0.95 power factor, 6% peak penetration, voltage, reactive-generation, and apparent-power limits are retained. Every OPF converges with zero voltage violation and at most 1.000-p.u. branch loading; the payment guarantee remains scoped to (23). The complete corrective panel adds 5,400 OPFs over all 6 IEEE-9 and 19 IEEE-14 non-islanding line outages, but does not claim a shared preventive plan. The distinct preventive panel fixes every intact non-reference active output across all six IEEE-9 outages as in (26). All 3,888 cells (54 days, four counterfactuals, and 3%/6%/9% penetration) converge with zero plan deviation and voltage violation; maximum loading is 0.876/0.899/0.923 p.u. This verifies the declared shared-active-plan claim without contingency screening. The intact, corrective, and shared-plan AC panels are retained in the experiment output.

Experiment 11 removes fixed trace placement and power scale by crossing all $4! = 24$ region-to-bus mappings, 3%, 6%, and 9% peak penetrations, four counterfactuals, and all 54 days, yielding 15,552 outcomes. The risk-constrained verifier reduces mean absolute payment error relative to the closest feasible-quantile projection from \$95.475814/\$191.816652/\$289.187860 to \$91.261724/\$183.245675/\$276.253993 per day, and is lower in all 72 mapping–scale cells. The public IEEE-118 native load is fixed at the predeclared 0.90 multiplier used by the N–1 stress panel, while native branch ratings remain unchanged. Maximum loading and locational marginal price spread are 1.000 p.u. and \$4.06/MWh. The corresponding panels separate the conversion-scenario cap, shared-active- plan AC envelope, and spatial-scale transfer from the base-case accuracy results, so these claims are not conflated with the trace-anchored mechanism isolation result. The complete assignment–penetration panel is retained in the experiment output.

TABLE IV
CONTINUOUS ROLLING-HORIZON MARKET VALIDATION OVER 54 LOCKED DAYS.

Metric	payment-certified verifier	Quantile	risk-constrained verifier	Single
Complete-cycle MAE (\$/event)	0.084761	0.053170	0.241926	0.084761
Event-only MAE (\$/event)	8.091153	11.655725	11.186565	10.763272
Max window residual (MW)		6.0×10^{-6}		
Max cycle-energy residual (MWh)		3.0×10^{-6}		
Verified capacity (MWh)	13.902	13.902	13.902	13.902
Positive contracts		54/54		

Experiment 12 removes the daily reset from the final market validation. Each locked event is embedded in 1,216 observed slots (608 pre-event, 96 event-day, and 512 recovery slots). Only the eight submitted event intervals are projected; all other decisions are endogenous. Across 216 method–day outcomes, the maximum event-window L1 residual is 6.0×10^{-6} MW and the maximum absolute cycle-energy residual is 3.0×10^{-6} MWh. Complete-cycle accounting gives a mean absolute remuneration error of \$0.084761 per event for payment-certified verifier, compared with \$0.053170 for the feasible-quantile reference, \$0.241926 for risk-constrained verifier, and \$0.084761 for the single projection; the event-only errors are respectively \$8.091153, \$11.655725, \$11.186565, and \$10.763272. At the predeclared site and \$150/MWh service price, mean verified capacity is 13.902285 MWh; all 54 transaction surpluses are positive (mean \$2,034.810894, minimum \$3.685837). Equation (21) therefore activates every contract with minimum participant/operator utility \$1.842918; maximum site and bilateral budget residuals are 1.42×10^{-14} and zero dollars. The Table IV condenses the complete-cycle accounting and residual bounds; the full time-resolved diagnostic is provided with the supplementary results.

D. Independent trace and deployment-boundary audits

The independent audit tier separates measured execution from the semi-synthetic intervention. Across all 54 locked days, replaying the immutable MIT scheduler/DCGM execution intervals reduces mean slot MAE from 6.615 to 4.287 MW and mean absolute energy discrepancy from 635.058 to 411.516 MWh, with 100% slot coverage. This is an observational fidelity check, not a causal utility-event estimate. A separate sparse job-level flow LP retains all 71,128 positive-energy joined jobs and 1,974,690 service variables; it reconstructs 2.657346 MWh with maximum job and aggregate-slot residuals of 6.0×10^{-18} and 9.4×10^{-18} MWh. The observed regional peak is 117.936 MW against the precommitted 118-MW nameplate. These results establish ledger fidelity and capacity coverage, not authenticity of untrusted semantic fields.

The held-out conversion-interval audit evaluates $q_{01} = 0.5845$ and $q_{99} = 2.6931$ with an independent 40-segment evaluator. Endpoint margins are \$0.545543/day and \$4.065188/day, with maximum cap violation 1.7×10^{-8} USD; Proposition 5 therefore certifies only the worst-case cost over the declared closed interval. The decision-time audit removes all post-gate arrivals and changes nRMSE from 0.348412 to 0.525448 and

TABLE V
INDEPENDENT REPLAY, INTERVAL-PAYMENT, AND LEDGER-PROVENANCE CERTIFICATES.

Evidence	Quantity	Baseline / value	Verified result
Trace replay	Mean slot MAE (MW)	Batch-only 6.615	Trace-constrained 4.287
Trace replay	Mean energy discrepancy (MWh)	Batch-only 635.058	Trace-constrained 411.516
Trace replay	Slot coverage	–	100% over 54 days
Flow witness	Jobs / service variables	–	71,128 / 1,974,690
Flow witness	Energy / maximum residual	–	2.657346 MWh / 9.4×10^{-18} MWh
Capacity	Observed peak / nameplate	–	117.936 / 118 MW = 0.9995
Interval certificate	Endpoint margins q_{01}/q_{99}	–	\$0.546 / \$4.065 per day
Interval certificate	Maximum cap violation	–	1.7×10^{-8} USD

credit recall from 0.389559 to 0.041674. Finally, 1,698 declared AC corrective recourse rows are solved across four public networks and three validation-scaled penetrations. These are model and information-boundary audits, not a preventive AC security-constrained OPF certificate or a field-event result.

VI. CONCLUSION

This paper develops a provenance-bound workload-state verifier and a security-aware nodal settlement for spatially coupled AI data centers. The cumulative state enforces release, deadline, conservation, migration, and capacity constraints; the two-sided credit band limits unsupported credited response and bounds additional under-credit; and signed N–1 value accounts for rebound and congestion. Across 54 locked days, false-credit exposure falls from 2.026218 to 1.910572 MWh/day relative to the matched feasible-quantile reference, while nRMSE falls from 0.353737 to 0.348412. The same comparison increases under-credit from 15.225 to 15.314 MWh/day and does not improve F_1, so the empirical result is a risk tradeoff rather than uniform predictive dominance. The trace-anchored mechanism-isolation panel reduces N–1 exact-settlement error from \$192.69 to \$0.90/day, whereas the end-to-end workload-verifier errors remain \$207.67, \$155.25, and \$155.40/day for the three settlement variants. The job-level witness preserves 71,128 measured jobs with machine-precision residuals, and the provenance certificate binds the witness to the immutable scheduler/DCGM join and reconciles the precommitted 118-MW nameplate with the observed regional envelope. The decision-time panel quantifies the information cost of removing post-gate arrivals, while the cross-network AC audit evaluates every declared native-case admissible outage on four public systems with validation-only power scaling. The public traces provide reproducible workload and network evidence rather than a utility field intervention. Deployment therefore requires a signed operator ledger, co-located facility telemetry, topology access, and a preapproved workload-to-power conversion; these requirements define the next intertemporal AC and utility-event validation stage.

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